AVAILABILITY MATH

Four nines is a budget, not a grade.

52.56

minutes of downtime a year at 99.99%

What the common availability targets actually allow.

Swipe for the full table →

THE ALLOWANCE

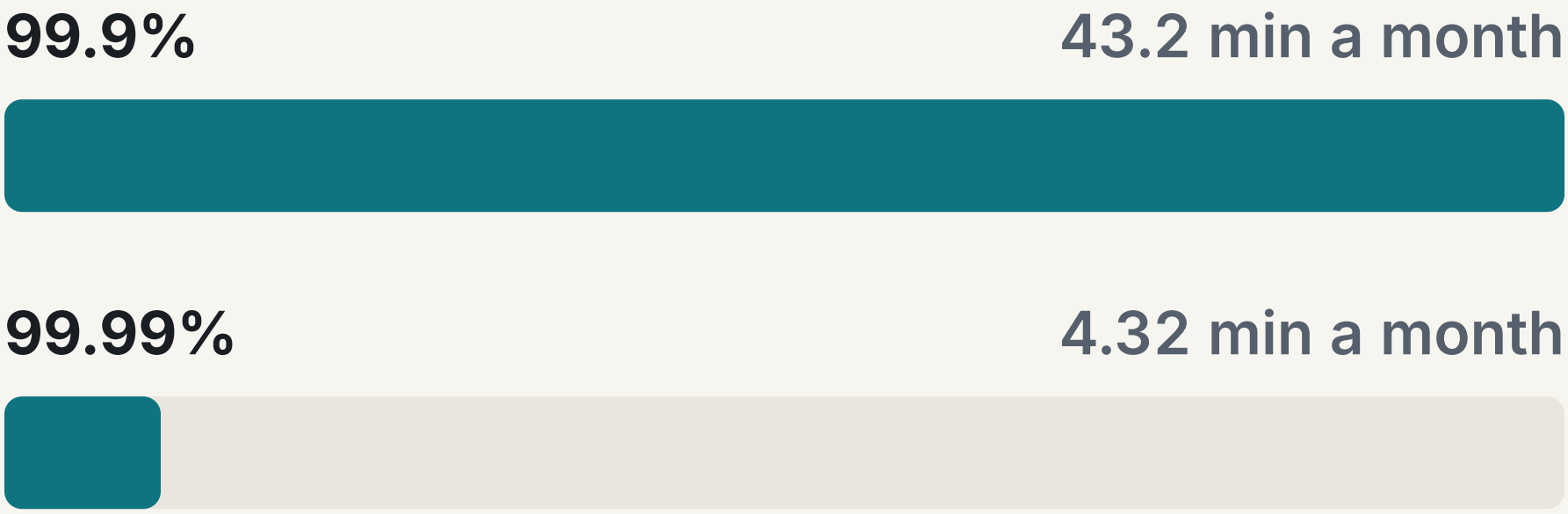
The downtime budget behind each target

TARGET	PER YEAR	PER 30-DAY MONTH
99%	87.6 hours	7.2 hours
99.9%	8.76 hours	43.2 minutes
99.95%	4.38 hours	21.6 minutes
99.99%	52.56 minutes	4.32 minutes
99.999%	5 min 15 s	26 seconds

Assumes a 365-day year and a 30-day month. Seconds are rounded.

EACH EXTRA NINE

Each nine leaves ten times less time

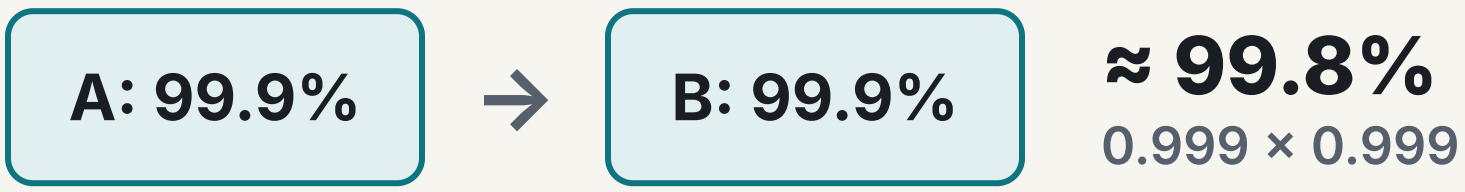


> At 99.99%, one 45-minute maintenance window uses about 86% of the whole year's allowance, before anything has broken.

DEPENDENCIES

Everything in the path shares the budget

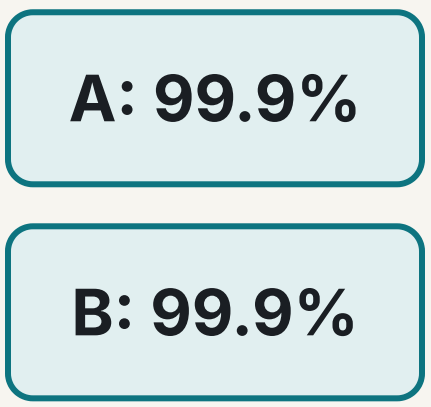
IN SERIES: BOTH MUST WORK



≈ 99.8%

0.999 × 0.999

IN PARALLEL: EITHER CAN SERVE



99.9999%

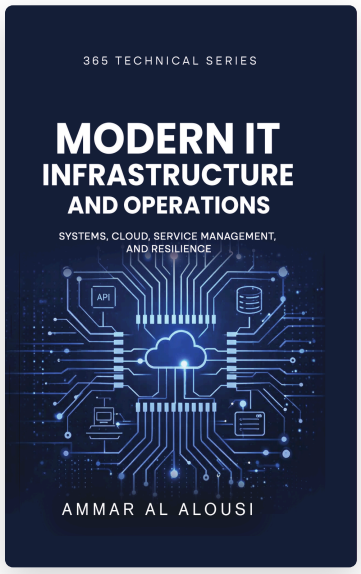
on paper: 1 − 0.001 × 0.001

Only if failures are independent. Shared power, network paths, configuration, or one deployment pipeline make them fail together.

BEFORE YOU COMMIT

Translate the target into minutes first

Convert it to minutes per month. Then list who spends them: maintenance, deployments, failures in services you depend on, and the incident nobody has had yet.



Go deeper on resilience in

Modern IT Infrastructure and Operations

Link in bio

52.56 minutes: Google SRE book, “Embracing Risk” (sre.google). Other figures calculated from a 365-day year and 30-day month; parallel figure assumes independent failures.